

Submission CALC1-FINAL-SUB06

Question CALC1-FINAL-Q01

notation changes halfway:

1. $f'(x) \approx 2x(3x-1)+(x^2+2)^3$
2. $f'(3) \approx 6(8)+(11)^3 \approx 15$

marks, subscripts, and units are mixed up

result written as: 15 (please interpret)

Question CALC1-FINAL-Q02

notation changes halfway:

1. $\int 3x \, dx \approx (3/2)x^2$
2. $[3/2 x^2]_0^5 \approx 37.5$

marks, subscripts, and units are mixed up

result written as: 37.5 (please interpret)

Question CALC1-FINAL-Q03

notation changes halfway:

1. $2w+2l \approx 20 \Rightarrow l \approx 10-w$
2. $A(w) \approx w(10-w)$; at $w \approx 7$, $A \approx 21$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (please interpret)

Question CALC1-FINAL-Q04

notation changes halfway:

1. $x^2-25 \approx (x-5)(x+5)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 5} x+5 \approx 10$

marks, subscripts, and units are mixed up

result written as: 10 (please interpret)

Question CALC1-FINAL-Q05

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up

result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)

Question CALC1-FINAL-Q06

notation changes halfway:

1. $f'(x) \approx 2x(3x-1)+(x^2+4)^3$
2. $f'(1) \approx 2(2)+(5)^3 \approx 11$

marks, subscripts, and units are mixed up
result written as: 11 (please interpret)

Question CALC1-FINAL-Q07

notation changes halfway:

1. $\int 3x \, dx \approx (3/2)x^2$
2. $[3/2 x^2]_0^2 \approx 6$

marks, subscripts, and units are mixed up
result written as: 6 (please interpret)

Question CALC1-FINAL-Q08

notation changes halfway:

1. $2w+2l \approx 14 \Rightarrow l \approx 7-w$
2. $A(w) \approx w(7-w)$; at $w \approx 4$, $A \approx 12$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up
result written as: $A(w)=w(7-w)$; $A(4)=12$; check $A'(w)=0$ (please interpret)

Question CALC1-FINAL-Q09

notation changes halfway:

1. $x^2-25 \approx (x-5)(x+5)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 5} x+5 \approx 10$

marks, subscripts, and units are mixed up
result written as: 10 (please interpret)

Question CALC1-FINAL-Q10

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up
result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)